

EXTENDED OPTIMIZATION ALGORITHM APPROACH FOR IMPROVING DATA CLUSTERING

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ABSTRACT

A review based on different types of clustering algorithms with their corresponding data sets has been proposed. In this paper, we have given a complete comparative statistical analysis of various clustering algorithms. Clustering algorithm usually employ distance metric or similarity metric to cluster the data set into different partitions. Well known clustering algorithms have been widely used in various disciplines. Type of clustering algorithm used comparatively easy to implement as data are invariably real number and can be used for statistical applications. Other type of data set such as categorical, time series, Boolean, spatial, temporal have limited applications. By viewing the statistical analysis, it is observed that there is no optimal solution for handling problems with large data sets of mixed and categorical attributes. Some of the algorithms can be but their performance degrades as size of data keeps in increasing.

DECLARATION BY SCHOLAR

I hereby declare that the research work reported in the dissertation entitled "EXTENDED OPTIMIZATION ALGORITHM APPROACH FOR IMPROVING DATA CLUSTERING" in partial fulfilment of the requirement for the award of Degree for Master of Technology in Computer Science and Engineering at Lovely Professional University, Phagwara, Punjab is an authentic work carried out under supervision of my research supervisor Dr. Sudhanshu Prakash Tiwari. I have not submitted this work elsewhere for any degree or diploma.

I understand that the work presented herewith is in direct compliance with Lovely Professional University's Policy on plagiarism, intellectual property rights, and highest standards of moral and ethical conduct. Therefore, to the best of my knowledge, the content of this dissertation represents authentic and honest research effort conducted, in its entirety, by me. I am fully responsible for the contents of my dissertation work.

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SUPERVISOR'S CERTIFICATE

This is to certify that the work reported in the M.Tech Dissertation entitled “**Extended Optimization Algorithm Approach For Improving Data Clustering**”, submitted by **Prabhat Kumar Pankaj** at **Lovely Professional University, Phagwara, India** is a bonafide record of her original work carried out under my supervision. This work has not been submitted elsewhere for any other degree.

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This period proved for me one of the most productive and knowledgeable experiences of my career. It provided me an opportunity to upgrade my skills as well as sharpen my professional knowledge and understanding in an environment that was intellectually stimulating.

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INTRODUCTION

Data cluster also be a troublesome task, whose downside is caused by an in-sufficiency of distinctive and precise definition of cluster. It's usually acknowledge that cluster is associate in nursing will posed disadvantage once previous knowledge concerning the underlying data distribution is not well made public ability to find clusters with varied shapes and sizes cloud also be a basic limitation of every cluster formulae in spite of cluster criterion(objective function) used. Discovery of majority or all of the clusters (of absolute shapes)gift at intervals the data cloud also be a long-standing goal of explorative pattern analysis. A framework is needed that integrates the outputs of multiple cluster algorithms, almost like varied cluster objectives, applied to identical data, to fulfil this goal.

Cluster could also be technique of assignment similar data into a same cluster or subsets (called clusters) therefore observations at intervals identical cluster unit similar in some sense. Cluster could also be a method of unattended learning, and a typical technique for applied mathematics data analysis utilized in many fields, machine learning. Processing pattern recognition, image analysis, knowledge retrieval, and bioinformatics.

1.1 Clustering

Clustering is data mining technique in which the data object are divided into the classes or clusters those have similar objects. This technique does not have pre-defined classes or clusters which make it different from classification in which class labels are defined. It is also considered as the most important unsupervised learning technique. The main motive or objective of cluster analysis is to find the group of objects in other group. Here, if the intra-cluster distances are minimum and inter-cluster distances is maximum then it forms good clusters.

There are various applications of clustering. Few of them are as follows:

- a) It is used in data analysis, text analysis, pattern recognition and image processing.
- b) It is also widely used by the companies to find out the patterns and discover therelationship between the products that they usually buy together and these relationship will help to increase the sale and profit directly increasing the marketing of the company.

- c) It is also used in the biology field in which the genes are categorized according to the features they have.
- d) It is also used for web analysis.
- e) It is also used by banks for detecting the credit card fraud by detecting the outliers.
- f) Clustering is also used for medical diagnostic.

1.2 Clustering algorithm techniques

Clustering algorithm are broadly classified in this various types:-

- a) Partitioning based = In this method, firstly the initial clusters are formed. Then the data is divided into different partitions where partitions are also known as clusters. In this algorithm each cluster must have at least one object and each object must belong to exactly one group. Different partitioning based algorithms are k-means [19], FCM [19].
- b) Hierarchical based = In this method, the whole data is organised in a hierarchical manner which has a root node and different leaf nodes. The leaf nodes represent the datasets. This clustering can be further of two types (either bottom-up or top-down). In bottom-up, it merges two or more clusters and in top-down approach, it splits the clusters into two or more clusters. The main disadvantage of this algorithm is that if the step is performed once on the dataset then it cannot be undone. Hierarchical based algorithm contains many different algorithms.
- c) Density based = In this method, the objects are divided according to different parameters like their density regions, connectivity and boundary. The main advantage of this algorithm category is that it can form clusters of arbitrary shapes and it also provides a way to handle the outlier values in a dataset. Different type of density based algorithms are DBSCAN, OPTICS, etc.
- d) Grid based = In this method, the objects are divided into different grids. The advantage of this algorithm is that it processes the whole dataset only once and finds the value for the grids which increases the processing time. This algorithm is based on the grid size and the size of grid is much less than that whole dataset because of which the performance of this category are Wave cluster, OPTIC grid, etc.
- e) Mode based = In this method, a model is hypothesized for each cluster to find the best fit of data for given model. The advantage of this method is that it also take care of

outliers and it also finds the initial number of clusters that are created. Statistical and neural networks are the two different approaches that are based on this clustering method. Different types of algorithms under this method.

1.3 K-means clustering

K-means [19] is one of the simplest unsupervised learning algorithm that solve the well-known clustering problem. The procedure of follows a simple and easy way to classify a given data set through a certain number of cluster (assume k clusters) fixed a prior. The main idea is to define k centre, one for each cluster. These centres should be placed in a cunning way because of different location causes different result. So, the better choice is to place them as much as possible far away from each other. The next step is to take each point belonging to given data set and associate it to the nearest centre. When no point is pending, the first step is completed and an early group age is done. At this point we need to re- calculate k new centroids as barycentre of the clusters resulting from the previous step. After we have these k new centroids, a new binding has to be done between the same dataset points and the nearest new centre. A loop has been changes their location step by step until no more changes are done or in other words centres do not move any more.

1.4 Asymmetric clustering

The basic criterion for statistical clustering is to maximize the resulting probability (or minimize the resulting perplexity) of the training data. Many traditional clustering techniques attempt to maximize the average mutual information of adjacent cluster where the same clusters are used for both predicated and conditional words. We will call these clustering techniques symmetric clustering, and the resulting clusters both clusters. In constructing the ACM [18], we used asymmetric clustering, in which different clusters are used for predicted and conditional words. In particular, for clustering conditional words, we try to minimize the perplexity of training data for bigram of the form.

The basic approach to clustering we used is top-down, splitting clustering algorithm. In each iteration, a cluster is split into two clusters in the way that the splitting achieves the maximal entropy decreased. Finally, we can also perform iterations of swapping all words between all clusters until convergence i.e. no more entropy decrease can be found. We find that our algorithm is much more efficient than agglomerative clustering algorithms, those which merge words bottom up.

1.5 Mean Shift

Mean shift is non-parametric feature-space analysis technique for locating the maxima of density function. Mean shift is procedure for locating the maxima of a given discrete data sampled from that function. It is useful for detecting the modes of this density. This is an iterative method, and we start with an initial estimate. Although the mean shift algorithm has been widely used in many applications, rigid proof for the convergence of the algorithm using a general kernel in high dimensional space is still missing. The convergence of the mean shift algorithm in one dimension with differentiable, convex and strictly decreasing profile function. However, the one dimensional case had limited real world applications. Also, the convergence of the algorithm in higher dimensions with a finite number of the (or isolated) stationary points has been proved. However, a sufficient condition for general function to have finite (or isolated) stationary points have not been proved.

In each iteration of the algorithm, is performed for all simultaneously. The first question, then, is how to estimate the density function given a sparse set of samples. One of the simplest approaches is to just smooth the data, e.g., by convolving it with a fixed kernel width, where the input samples and kernel function (or Parzen window). This approach is known as kernel density estimation or the Parzen window technique. Once we have performed from the equation, we can find its local maxima using gradient ascent or some other optimization literature

1.6 Affinity metrics

Since the principle of proximity and smooth continuation arise from local properties of the configuration arise from local properties of the configuration of the edges, we can model them using only local information. Both of these local properties are modeled by the distribution of smooth curves that pass through two given edges. The distribution of curves is modeled by a smooth stochastic motion of a particle. Given two edges, we determine the probability that a particle starts with the position and direction of the second edge. The affinity from the first to the second edge is the sum of the probabilities of all paths that a particle can take between the two edges. The change in direction of the particle over time is normally direction of the particle over time is normally distributed with zero mean. Smaller the variance of the distribution, the smoother are the more probable curves that pass between two edges. Thus the variance of the normal distribution models the principle of smooth continuation. In addition each particle has a non-zero probability for decaying at anytime. Hence, edges that are farther apart are likely to have fewer curves that pass through

both of them. Thus the decay of the particle models the principle of proximity. The affinities between all pairs of edges form the affinity metrics.

1.7 MCL

The MCL is short for the markov cluster algorithm, a fast and scalable unsupervised cluster algorithm for graphs (also known as networks) based pm simulation of (stochastic) flow in graphs. The basic interface to the algorithm is very simple you need only one option (the `-I` flag) to get to the heart of it, and for large graphs you should also be aware of the `scheme` flag for regulating resources. The default approach is to vary the argument to `-I` over some interval (doing an MCL run for each value), and analyse the clustering output with the other programs that come with MCL.

MCL algorithm and process you are advised to read one of the technical reports among the publications. It is also possible to view a somewhat longer introduction or a possible to some of the mathematics associated with MCL.

1.8 S-clustering

Spectral clustering refers to a class of techniques which rely on the eigen-structure of a similarity matrix to partition points into disjoint clusters with points in the same cluster having high similarity and points in different clusters having low similarity and points in different clusters have low similarity. A new cost function for spectral clustering based on measure of error between a given partition and a solution of the spectral relaxation of a minimum normalized cut problem. Minimizing this cost function with respect to the partition leads to a new spectral clustering algorithm. Minimizing with respect to the similarity matrix leads to an algorithm for learning the similarity matrix. We develop a tractable approximation of our cost function that is based on the power method of computing eigenvectors.

1.9 PSO

Particle swarm optimization might sound complicated, but it's really a very simple algorithm. Over a number of iterations, a group of variables have their values adjusted closer to the member whose value is closest to the target at any given moment. Imagine a flock of birds circling over an area where they can smell a hidden source of food. The one who is closest to the food chirps the loudest and the other birds swing around in his direction. If any of the other circling birds comes closer to the target than the first, it chirps louder and the others veer over toward him. The tightening pattern continues until one of the

birds happens upon the food. It's an algorithm that's simple and easy to implement. The algorithm keeps track of three global variables- target value or condition, global best (gbest) value indicating which particle's data is currently closest to the target, stopping value indicating when the algorithm should stop if the target isn't found. Each particle consists of- data representing a possible solution, a velocity value indicating how much the data can be changed, a personal best (pbest) value indicating the closest particle's data has ever come to the target.

1.10 ACO

Ant colony optimization is a population-based meta-heuristic that can be used to find approximate solution to difficult optimization problems. The optimization problem is transformed into the problem of finding the best path on weighted graph. The artificial ants incrementally build solutions by moving on the graphs. The solution construction process is stochastic and is biased by a pheromone model, that is, a set of parameters associated with graphs components whose values are modified at runtime by the ants. The ants construct the solutions as follow. Then at each construction step it moves along the edges of the graph. Each ant keeps a memory of its path, and in subsequent step it chooses among the edges that do not lead to vertices that it has already visited. The probabilistic rule is biased by pheromone values and heuristics information: higher pheromone and the particular edge. Once all the ants have completed their tour, the pheromone on the edge is updated. Each of the pheromone values is initially decreased by certain percentage. Each edge then receives an amount of additional pheromone proportional to the quality of the solutions to which it belongs (there is one solution per ant).

CHAPTER 2

REVIEW OF LITERATURE

Ghali, et.al (2009) [1] “Exponential Particle Swarm Optimization Approach for Improving Data Clustering” use exponential particle swarm optimization (EPSO) to cluster data. Then we compare between (EPSO) clustering algorithm which depends on exponential variation for the inertia weight and particle swarm optimization (PSO). EPSO and PSO clustering are applied on five datasets, respectively. The Euclidian distance measure is used as the similarity metrics in each algorithm. For any easy comparison, ESPO and PSO approaches run 1000 iterations in each experiment. For all the result reported, averages over 30 simulations are given. The results obtained from the two clustering algorithms for the problems above. First, consider the fitness of solutions, i.e. the quantization error. For all problems, the EPSO algorithm had the smallest average quantization error in all problems, while the PSO algorithm had a large quantization error it is also illustrates that PSO fail to reach the optimal minimum in some runs, but the EPSO successes to reach the optimal minimum in all runs for all problems except for Iris plants database problem has lessnumber of failures, we notice that EPSO reaches best minimum points better than the liner PSO. The application of the EPSO to cluster vectors. The EPSO algorithm was compared against the PSO clustering algorithm which showed that the EPSO convergence slower to lower quantization error, while the PSO convergence faster to large quantization error. Also the proposed EPSO increases the possibility to find the optimal positions as it decrease the number of failure.

Salima Quadfel, et.al (2010) [2] “A modified particle swarm optimization algorithm for automatic image clustering” automatic image clustering algorithm based on a modified version of particle swarm optimization algorithm. ACMPSO clustering algorithm can partition image into compact and well separated clusters without any knowledge on the real number clusters. It uses a swarm of particles with variable number of length, which evolve dynamically using mutation operators. To evaluate the ability of our algorithm ACMPSO to find the correct number of clusters, we have tested it using natural images with varying range of complexity. The performance of ACMPSO, is compared with the recently developed automatic clustering algorithms known as the DCPSO[6], the VGA[6] and the ISODATA[3]. A new particle swarm optimization based method for automatic image clustering. The proposed algorithm uses a swarm of particles with variable length to deal

with the unknown number of clusters. Each particle encoded a partition of image with a number of clusters chosen randomly from the set of the maximum number of clusters. The partition of each particle of the swarm evolves using mutation operators which aim to reduce dynamically the number of clusters centroids. The proposed ACMPPSO algorithm performed very well when compared to other state-of-the-art clustering algorithm.

F.Samadzadegan, et.al (2009) [3] “Clustering of LIDAR data using particle swarm optimization algorithm in urban area” the transformation of the LIDAR data into the meaningful objects in urban area involves their segmentation into consistent units through a clustering process. Nevertheless, due to the scene complexity and the variety of the objects in urban area, e.g. buildings, roads, and trees, it is clear that a clustering using only a single cue will not suffice. Considering the availability of additional data source, like laser range and intensity information in both first and last echo, more information can be integrated in the clustering process and ultimately into the recognition and reconstruction scheme. Multi dimensionality nature of LIDAR data with a dense sampling interval in urban area generates a huge amount of information. The data are recorded in area of Rheine in Germany. Two different patches with residential and industrial patten were selected in the available dataset. The selected areas were suitable for the evaluation of the proposed classification strategy because the required complexities were available in the image. The capability of new clustering method based on the particle swarm optimization algorithm in object extraction from LIDAR data. The method employs the PSO algorithm to search for the set of clusters centres that minimizes a given clustering metric. One of the advantages of this method is that it does not become trapped at locally optimal solutions. This is due to ability of the PSO algorithm to perform local and global search simultaneously.

Xiao Cai, et.al (2013) [4] “Multi-View K-Means Clustering on Big Data” more and more data are collected from multiple sources or represented by multiple views, where different views describe distinct perspectives of the data. Although each view could be individually used for finding patterns by clustering, the clustering performance could be more accurate by exploring the rich information could be more accurate by exploring the rich information among multiple views. Several multi-view clustering methods have been proposed to unsupervised integrate different views of data. However, they are graph based approaches, e.g. based on spectral clustering, such that they cannot handle the large-scale data. How to combine the heterogeneous features for unsupervised large scale data clustering has become challenging problem. We will compare the multi-view clustering performance of our

method with their corresponding single-view counterpart. In addition, we also compare the results of our method with the baseline method naïve multi-view K-means clustering and affinity propagation. In our method, when we ignore the weight learning for each type of visual features, the method degenerates to a simple version, called as simple MKMC. Utilizing the common cluster indicator, we can search a consensus pattern and do clustering across multiple visual feature views. Moreover, by imposing the structured sparsity ℓ_1 -norm on the objective function, our method is robust to the outliers input data. Our new method learns the weights of each view adaptively.

Mehdi Neshat, et.al (2012) [5] “A New Cooperative Algorithm Based on PSO and K-means for Data Clustering” data clustering has been applied in multiple fields such as machine learning, data mining, wireless sensor networks and pattern recognition. One of the most famous clustering approaches is K-means which effectively has been used in many clustering problems, but this algorithm has some drawbacks such as local optimal convergence and sensitivity to initial points. The particles are initialized in PSO. Each of the particles are initialized in PSO. Each of particles contain K cluster centers which are displaced in the problem space by performing PSO algorithm. PSO continues to perform until the particles converge. After convergence of PSO, Gbest position including the best cluster centers which have found by particles so far is considered as the input of k-means. A new cooperative algorithm based on K-means and PSO is presented. In the proposed algorithm, PSO performs global search and k-means is responsible for local search. The process of the proposed algorithm is such that the strength and ability of preventing from being trapped in local optimums is improved. The proposed algorithm along with four other algorithms is used for clustering of datasets.

Sandeep Rana, et.al (2010) [6] “A hybrid sequential approach for data clustering using K-means and particle swarm optimization algorithm” most clustering algorithms belong to two groups: hierarchical clustering and partitioned clustering. The hierarchical approach produces a nested series of partitions consisting of clusters either disjoint or included one into the other. The application of PSO in sequence with k-means to clustering problem. Five algorithms are tested namely a standard k-means, PSO, K-means + Genetic algorithm, hybrid approach and the hybrid sequential clustering algorithm where the swarms find the cluster centre and further refining is obtained through k-means algorithm. The hybrid sequential clustering algorithm is compared with two PSO approaches, K-Means clustering and with Genetic algorithm, which shows that the proposed clustering algorithm have better

convergence to lower quantization errors, and in general, larger inter-cluster distances and smaller intra-cluster distance. The variation in the solutions obtained for different cases is also reported minimum in the proposed algorithm. It can be concluded that the drawback of finding optimal solution by K-means can be minimized by using PSO over it. The variations in PSO algorithm and its hybridization with K-Means algorithm is proposed for future research.

Manjunath, et.al (2012) [7] “Realistic Analysis of data ware hosing and data-mining application in education domain” data driven support systems, such as data warehouses can serve the requirement of extraction of information from more than one subject area. Data warehouses can provide the information required by the decision makers. Developing a warehouse for educational institute is the less focus area since educational institute are non-profit and service oriented organizations. In present day scenario where education has been privatized and cut throat competition is prevailing, institute needs to be more organized and need to take better decisions. This paper have explored the need of data warehouses / business intelligence for an educational institute, the operational data of an educational institution has been used for experimentation. Benefits of the project can be more if the institute has the positive approach towards new technologies. They can take micro-level decisions in a timely manner without the need to depend on their IT staff. They can perform extensive analysis of stored data to provide answers to the exhaustive queries to the administrative cadre. This help them to formulate strategies and policies for employees and student. This helps student and Employees in making decisions. They are the ultimate beneficiaries of the new policies formulated by the decision maker and policy planner’s extensive analysis on student and employee related data.

Marek Smaieja, et.al (2014) [8] “Asymmetric Clustering Index in a case of 5-HT_{1A} Receptor Ligands” rapidly growing number of compounds with a determined activity for a given molecular target leads to difficulties in using full, previously explored chemicals spaces in virtual screening campaign spaces in virtual screening campaigns. Indeed, the use of a large number of ligands (e.g., the D₂ receptor has 9180 different ligands in ChEMBL database v.16) in predictive model development usually generates substantial computational costs. Manual (knowledge based) clustering is usually the first choice for small groups of ligands because it provides the most natural partitions. Several methods have been proposed to compare clustering. The most popular techniques are based on counting pairs of element classified in the same way in both partitions, such as the rand index and its modification.

Another group of methods uses normalized mutual information to quantify the information shared by the clustering. An interesting approach for comparing partitions relies on measuring the distance between clustering with the use of information theory. The main feature of these indices is their symmetry, which makes them suitable for finding the similarities between clustering. The Asymmetric index, which allows one to evaluate how well a numerically constructed partition reflects the reference. The highest ACI was consistently obtained for hierarchical clustering based on the complete linkage function, the Kelkote-Roth fingerprint and the Buser similarity metric, suggesting the application of these parameters for other groups of biologically active compounds.

Law, et.al (2004) [9] “Multiobjective data clustering” Conventional clustering algorithms Utilize a single criterion that may not conform to the diverse shapes of the underlying clusters. We offer a new clustering approach that uses multiple cluster objective functions simultaneously. The proposed multi objective clustering is a two-step process. Includes detection of clusters by a set of candidate objective functions as well as their integration into the target partition. A key ingredient of the approach is a cluster goodness function that evaluates the utility of multiple clusters using re-sampling techniques. Multiobjective data clustering is obtained as a solution to a discrete optimization problem in the space of clusters. The goal of multi objective clustering is to find clusters in a data set by applying several clustering algorithms corresponding to different objective functions. We propose a clustering approach that integrates the output of different clustering algorithms into a single partition. More precisely, given different clustering objective functions, we seek a partition that utilizes the appropriate objective functions for different parts of the data space. The proposed method is based on the principle of data space partitioning, where different learning algorithms are applied to different part of the space. There are two types of parameters in our approach. The parameters of the candidate clustering algorithms should be set in such a way that the algorithm have a reasonable chance to detect the underlying clusters.

Chuang, et.al (2012) “An Improved Optimization for data clustering” [10] clustering is still a popular analysis tool for data statistics. The data structure identifying from the large-scale data has become a very important issue in the data mining problem. In this paper, an improved particle swarm optimization based on Gauss chaotic map for clustering is proposed, Gauss chaotic map adopts a random sequence with a random starting point as a parameter, and relies on this parameter to update the positions and velocities of the particles. It provides the significant chaos distribution to balance the exploration and exploitation

capability for search process. This easy and fast function generates a random seed processes, and further improve the performance of PSO due to their unpredictability. PSO adopts a random sequence with a random starting point as a parameter, and relies on this parameter to update the positions and velocities of the particles. However, PSO often leads to premature convergence, especially in complex multi-peak search problems such clustering of dimensional. We combined the Gauss chaotic Map and high-dimension particle swarm optimization, named GaussPSO. Results of the conducted experimental trials on a variety of data sets taken from several real-life situations demonstrate that proposed Gauss PSO is superior to the K-means, PSO, NM-PSO, K-PSO, and K-NM-PSO algorithms.

Seetha, et.al (2012) [11] “An efficient hybrid particle swarm optimization for data clustering” an efficient hybrid method, namely fuzzy particle swarm optimization (FPSO) and fuzzy c-means (FCM) algorithms, to solve the fuzzy clustering problem, especially for large sizes. When the problem becomes large, the FCM algorithm may result in uneven distribution of data, making it difficult to find an optimal solution in reasonable amount of time. The PSO algorithm does find a good or near optimal solution in reasonable time, but its performance was improved by seeding the initial swarm with the result of the c-means algorithm. The main objective of this study is to assess the relative performance of the proposed FPSO with respect to the FCM modification. The performance is measured by the objective function value and the CPU execution time. The objective function value is analyzed for FCM and proposed FPSO by varying the number of clusters for both the datasets (i.e., Iris and Zoo datasets). An analysis is done to find the better algorithm between FCM and FPSO using OFV with increased number of clusters. Lower the OFV for an algorithm, the better the clustering. If time taken by any algorithm to complete the execution is less than that algorithm can be considered to be best. Fuzzy clustering is a process of assigning these membership levels, and then using them to assign data elements to one or more clusters. In real-world cases, there may be no sharp boundaries between clusters, and in such cases, fuzzy clustering will be a better choice for the data.

Zheng, et.al (2015) [12] “An improved PSO clustering algorithm with entropy-based fuzzy clustering” Particle swarm optimization is a based population heuristic global optimization technology and is referred to as a swarm intelligence technique. In general, each particle is initialized randomly which increases the iteration time and makes the result unstable. In this paper an improved clustering algorithm combined with entropy based fuzzy clustering (EFC) is presented. Firstly EFC gets an initial lute, then the cluster center is regarded as

inputs of or of all particles instead of being initialized randomly. Finally we cluster with the improved clustering algorithm which guarantees unique clustering. The experimental results show that the improved clustering algorithm has not only high accuracy but also certain stability. This section gives relevant performance of different clustering algorithms. The performance of PSO algorithm is better by initializing cluster center using EFC algorithm. Relevant experiments of K-means, the integration of K-means and PSO (KPSO), PSO and EFCPSO algorithms have been conducted on Iris and Wine datasets 10 times. The clustering problem can be regarded as a kind of particular global optimization problem. Hence the PSO algorithm which solve the optimization problem can also be used to cluster data.

Peter, et.al (1999) [13] “Mean Shift Analysis and Applications” a nonparametric estimator of density gradient, the mean shift, is employed in the joint, spatial-range(value) domain of grey level and colour images for continuity preserving filtering and image segmentation. Properties of the mean shift are reviewed and its convergence on lattices is proven. The proposed filtering method associates with each pixel in the images the closest local mode in the density distribution of the joint domain. Segmentation into a piecewise constant structure requires only one more step, fusion of the regions associated with nearby nodes. The proposed technique has two parameters controlling the resolution in the spatial and range domains. The nonparametric estimation of the density gradient in the spatial range domain is a useful tool for bottom-up computer vision tasks such as edge preserving filtering and segmentation. The methods proposed can be easily extended to the processing of other low level image features like the texture or optical flow.

Comaniciu et.al (2000) [14] “Real-Time Tracking of Non-Rigid Objects using Mean Shift” new method for real time tracking of non-rigid objects seen from a moving camera is proposed, the central computational module is based on the mean shift iterations and finds the most probable target position in the current frame. The dissimilarity between the target model (its colour distribution) and the target candidates is expressed by the metric derive from the Bhattacharyya coefficient. The theoretical analysis of the approach shows that it relates to the Bayesian framework while providing a practical, fast and efficient solution. The capability of the tracker to handle in real time partial, significant clutter and target scale variations, is demonstrated. The sample mean shift, introduce the iterative mean shift procedure, and present a new theorem showing the convergence for kernels with convex and monotonic profiles.

Shimshoni et.al (2003) [15] “Mena Shift Based Clustering in High Dimension: A texture classification Example” feature space analysis is the main module in many computer vision tasks. The most popular technique, k-means clustering, however, has two inherent limitations: the clusters are constrained to be spherically symmetric and their number has to be known a priori. In nonparametric clustering methods, like the one based on mean shift, these limitations are eliminated but the amount of computation becomes prohibitively large as the dimension of the space increases. We exploit a recently proposed approximation technique, locality-sensitive hashing (LSH), to reduce the computational complexity of adaptive mean shift. The k-means clustering imposes rigidly a given number of identical spherical clusters over the feature space. Thus it is expected that when this structure is not.

Grenier et.al (2010) [16] “Optimization of mean-shift scale parameter on the EGEE grid” the optimization of Mean-Shift (MS) image filtering scale parameter. A parameter sweep experiment representing 164 days of CPU is performed on the EGEE grid. The mathematical foundation of Mean-Shift and the grid environment used for the deployment are described in details. The experiments and results are then discussed highlighting the efficiency of gradient ascent algorithm for MS parameter optimization and a number of grid observation related to data transfers, reliability, task scheduling, CPU time and usability. The EGEE grid allowed to compute a 164-day experiment in 2.5 days, i.e., it speed up the experiment by a factor of 66. Mean-Shift filtering quality is highly dependent on scale parameters.

Werner Bailer, et.al (2005) [17] “Optimized mean shift algorithm for color segmentation in image sequences” apply mean shift color segmentation to image sequences, as the first step of a moving object segmentation algorithm. Evaluation of the segmentation results to compare the original algorithm with our modified version. We show that our optimized algorithm reduces processing time and increase the temporal stability of the segmentation. The mean shift algorithm is well suited for color segmentation of image sequences, as it provides better results than comparable algorithms. An optimized mean shift algorithm for color segmentation of image sequences. We follow the idea of using the resulting cluster centers of a frame as initialization for the subsequent frame in order to exploit the temporal redundancy and reduce computation time by 20-50%. The originally proposed CIE LUV color space has been replaced in other works for performance reason. The temporal stability of our algorithm is higher, especially in scenes with fairly low visual activity. We have also

shown that moderate quantization of the input data has no significant effect on the quality of the resulting segmentation, but speeds up computation by over 40%.

Maurizio Filippone, et.al (2008) [18] “A survey of kernel and spectral methods for clustering” partitioning clustering problem with special interest in two recent approaches: kernel and spectral methods. A survey of kernel and spectral clustering methods, two approaches able to produce nonlinear separating hypersurfaces between clusters. Spectral and kernel methods for clustering have been reviewed paying attention to fuzzy kernel for clustering and to the connection between kernel and spectral approaches. Unlike classical partitioning clustering algorithms they are able to produce nonlinear separating hypersurfaces among data since they construct an adjacency structure from data. These methods have been successfully tested on several benchmarks, but we can find few applications to real world problem due to the high computational cost. Therefore an extensive validation on real world applications remains a big challenge for kernel and spectral methods.

Francis R.Bach, et.al (2004) [19] “Learning Spectral Clustering” drive new cost function for spectral clustering based on a measure of error between a given partition and a solution of the spectral relaxation of a minimum normalized cut problem. Minimizing this cost function with respect to the partition leads to a new spectral clustering algorithm. One for spectral clustering and one for learning the similarity matrix. These algorithms can be derived as the minimization of a single cost function with respect to its two arguments. This cost function depends directly on the eigenstructure of the similarity matrix. We have shown that it can be approximated efficiently using the power method, yielding a method for learning similarity matrices that can cluster effectively in cases in which non-adaptive approaches fail. Our new approach yields a spectral clustering method that is significantly more robust to irrelevant features than current methods, and also speech separation and image segmentation, in particular with the objective of selecting features from among the numerous features.

M.P.S Bhatia, et.al (2013) [20] “Experimental study of Data clustering using K-means and modified algorithms” discuss about two modified approaches of k-means like number of cluster must be known in advance, does not deal with outliers and converges to local minima. The first modified approach of k-means clustering, the data set is partitioned into $k \times k$ segments where k is the number of clusters. Then threshold distance value which is denoted as $dc(i)$. If its value is less than or equal to threshold distance then the object is

assigned to that cluster else is less than or equal to threshold distance then the object is assigned to that cluster else the distance is again calculated from other centroids and this process continues unless the data is not assigned to a cluster. This approach is divided into two phase. In first phase, the initial centroids are determined and then in second phase points are assigned in the clusters. The datasets used to analyse the results are iris having 4 attributes and 150 instances. The result of analysis is evaluated on the basis of certain metrics and the metrics used by the authors to evaluate the result ere number of iterations, accuracy of clusters formed, and validity index used was silhouetee and execution time. The result of modified approaches showed better results than original k-means and also the first modified approach.

Derpanis, et.al (2005) [21] “Mean Shift Clustering” the mean shift clustering is a nonparametric clustering techniques which does not require prior knowledge of the number of clusters, and does not constrain the shape of the clusters. The mean shift clustering algorithm is practical application of the mode finding procedures are – starting on the data points, run mean shift procedure to find the stationary points of the density function, prune these points by retaining only the local maxima. The set of all location that converge to the same mode defines the basin of attraction of that mode. The points which are in the same basin of attraction is associated with the same cluster.

Dhote, et.al (2013) [22] “Swarm Intelligence techniques for Optimization in data clustering” clustering represents the large datasets by a structured well defined number of clusters or prototypes. K-means is a useful technique to data clustering which partitions the data into K-Clusters. However, the results of k-means algorithm are based on the selection of initial seeds and converge to local optimum solution. The Swarm Intelligence (SI) is an algorithm to apply many simple agents behaviour which inn terms lead to an emergent global behaviour solution. Data mining tasks require fast and accurate partitioning of large datasets, which may come with a number of attributes or features. The swarm intelligence optimization technique like ACO and PSO has successfully been applied to a number of realworld clustering problems in order to meet the clustering requirement.

Rezaei, et.al (2015) [23] “ A Hybrid Model of Particle Swarm Optimization and continues ant colony optimization for multimodal functions optimization” PSO particle swarm optimization and ACOR ant colony optimization are two population-based optimization techniques that work based on probability laws. The main problem of PSO and ACOR algorithm is premature convergence and falling into local optima. One way to solve the problems is to use combinational methods. This paper presents a combinational method including PSO and ACOR in order to improve the search process. The proposed algorithm tries to solve the problem. Standard benchmark functions are used in order to evaluate the proposed algorithm, proposed method was compared with ACOR.PSO and sequential

approach with the enlarge pheromone-particle table of the composition of PSO and ACOR. Result shows that the proposed method is superior.

Venkatesan, et.al (2011) [24] “Clustering of datasets using PSO-K-Means and PCA-K-Means” cluster analysis plays indispensable role in obtaining knowledge from data, being the first step in data mining and knowledge discovery. The purpose of data clustering is to reveal the data patterns and gain some initial insights regarding data distribution. K-means is one of the widely used patrimonial clustering algorithm and it is more sensitive to outliers and do not work well with high dimensional data. In this paper, basic K-means has been integrated with other approaches to overcome the shortcomings hereby improving the accuracy of clustering. Basic K-means and the combination of K-means with PCA and PSO are applied on various datasets from UCI repository. Results of this paper show that PSO-K-Means and PCA-K-Means improves the performance of basic K-means in terms of accuracy and computational time.

Zhong, et.al (2012) [25] “A Hybrid particle Swarm Optimization Algorithm based on space transformation Search and a Modified Velocity Model” Particle Swarm Optimization (PSO) has shown its fast search speed in many complicated optimization and search problems. However, PSO often easily falls into local optima because the particles would quickly get closer to the best particle. Under their circumstances, the best particle could hardly be improved. This paper proposes a new hybrid PSO (HPSO) to solve this problem by computing space transformation search (STS) with a new modified velocity model. Experimental studies on 8 benchmark functions demonstrate that the HPSO holds good performance in solving both unimodal and multimodal functions optimization problems.

Abbas, et.al (2008) [26] “Comparison between Data Clustering Algorithm” clustering is a division of data into groups of similar objects. Each group called a cluster consist of objects that are similar between themselves and dissimilar compared to objects of other groups. This paper is intended to study and compare different data clustering algorithms. The algorithm under investigation are K-means algorithm, hierarchical clustering algorithms, self-organizing maps algorithm, and expectation maximization clustering algorithms All these algorithms are compared according to the following factors : size of dataset, number of clusters, type of dataset and type of software used. Some conclusions that are extracted belong to the performance, quality and accuracy of the clustering algorithm. Running the clustering algorithm using any software gives almost the same results even when changing any of the factors because most software use the same procedures and ideas in any algorithm implemented by them.

Dahiya, et.al (2014) [27] “A survey on application of particle swarm optimization in text mining” Mining is the discovery of new previously unknown information, by automatically extracting information from a usually large amount of data set Many algorithms have been developed in recent years for solving problems of numerical and combinatorial optimization problems. Most efficient among them are swarm intelligence algorithms. Clustering using PSO is being used as an alternative to more convention clustering techniques. PSO is population-based stochastic search algorithms that impersonate the capability of swarm. Data clustering with PSO algorithms are being used to produce better outcomes in a wide

variety of real-world data. This paper has presented a review of previous researches conducted in the areas of Particle swarm optimization (PSO). PSO hybrids and their application to data clustering. Researches in this field shows that if PSO is hybridized with other clustering algorithms, then it yields better results in various optimization problems in terms of efficiency and accuracy when compared with other evolutionary algorithms such as GA, SA etc. The implementation of hybrid PSO algorithms for data clustering yields optimal number of clusters which results in better prediction and analysis of data. A comprehensive survey of literature in this area has therefore been given to help provide more insight in this subject.

Marie, et.al (2013) [28] “A PSO-Based Subtractive Data Clustering Algorithm” there is a tremendous proliferation in the amount of information available on the largest shared information source the World Wide Web. Fast and high quality clustering algorithms play an important role in helping users to effectively navigate, summarize, and organise the information. Recent studies have shown that partitioned clustering algorithm such as the k-means algorithm are the most popular algorithm for clustering large datasets the major problem with partitioned clustering algorithm is that they are sensitive to selection of initial partitions and prone to premature converge to local optima. Subtractive clustering is fast, one pass algorithm for estimating the number of clusters and cluster centers for any given set of data.

CHAPTER 3

PRESENT WORK

This chapter includes problem formulation, objectives of the following study, understanding the scope and research methodology of the work. The first subsection is talks about the problem around which whole of this study revolves. In second section, the objective and reason of this study and work is mentioned and lastly the actually methodology on which my work is based on is described. It includesthe flow chart which explains how the work is carried out.

3.1 PROBLEM FORMULATION

Clustering is also referred as unsupervised grouping of objects in which the labels of the groups are unknown. Clustering was done using the k-means algorithm which is an iterative process in which the distance is calculated iteratively. The K-means algorithm that was used resulted in improper clustering as the clusters that were formed were not linearly separable. After K-means algorithm Asymmetric algorithm was employed for clustering. In this clustering approach there were chances that two closely spaced points are located in separate clusters. So to overcome this drawback another algorithm was defined. Then, a new technique was developed that was Affinity Metric. In this technique the standard deviation of all the parameters is calculated and also a Euclidian distance value of standard deviation is set. Affinity algorithm is applied on only those parameters that have value of standard deviation lesser than the Euclidian distance set. This improved accuracy but had drawback like high computation time. In this technique the numbers of clusters are to be known in advance. This technique also did not work well with poor availability of the data. A new technique needs to be proposed that can overcome these drawbacks and clustering can be obtained with higher accuracy even with poor availability of data.

The conventional techniques that were used for optimization did not generated desired results for clustered data and the results of clustering were not that accurate. In the technique that is adopted in this work, the clustering is done using the Mean Shift Algorithm because it is considered to produce better and accurate results for clustering than Affinity Metric algorithm. It is considered that if the result of previous one

algorithm is get optimized with new techniques of algorithms then proposed algorithm is acceptable of providing better result for performance.

3.2 OBJECTIVE OF THE STUDY

The main objectives of this study are listed below:

1. It improves the accuracy of the existing algorithm.
2. It is reducing the noise.
3. It is also minimizes its overall time consumption and its iteration also.

3.3 RESEARCH METHODOLOGY

The methodology of the approach adopted in this study is described below:

- 1) The first step will be the select the datasets.
- 2) Load the dataset and identify the number of rows and columns of the datasets.
- 3) Now, the next step will be process the datasets and calculate the weight on the basis of relation between each element of the datasets.
- 4) If the weight calculated is maximum means its near to cluster otherwise again process the datasets and calculate the weight on the basis of relation between each element of the datasets and referred generated data.
- 5) Define central point in generated data matrices.
- 6) Calculate the similarity using affinity matrix from define central point in generated data.
- 7) S-Cluster is assign to each point of generated data.
- 8) Finally data is assigned the clusters.

3.4 FLOW CHART

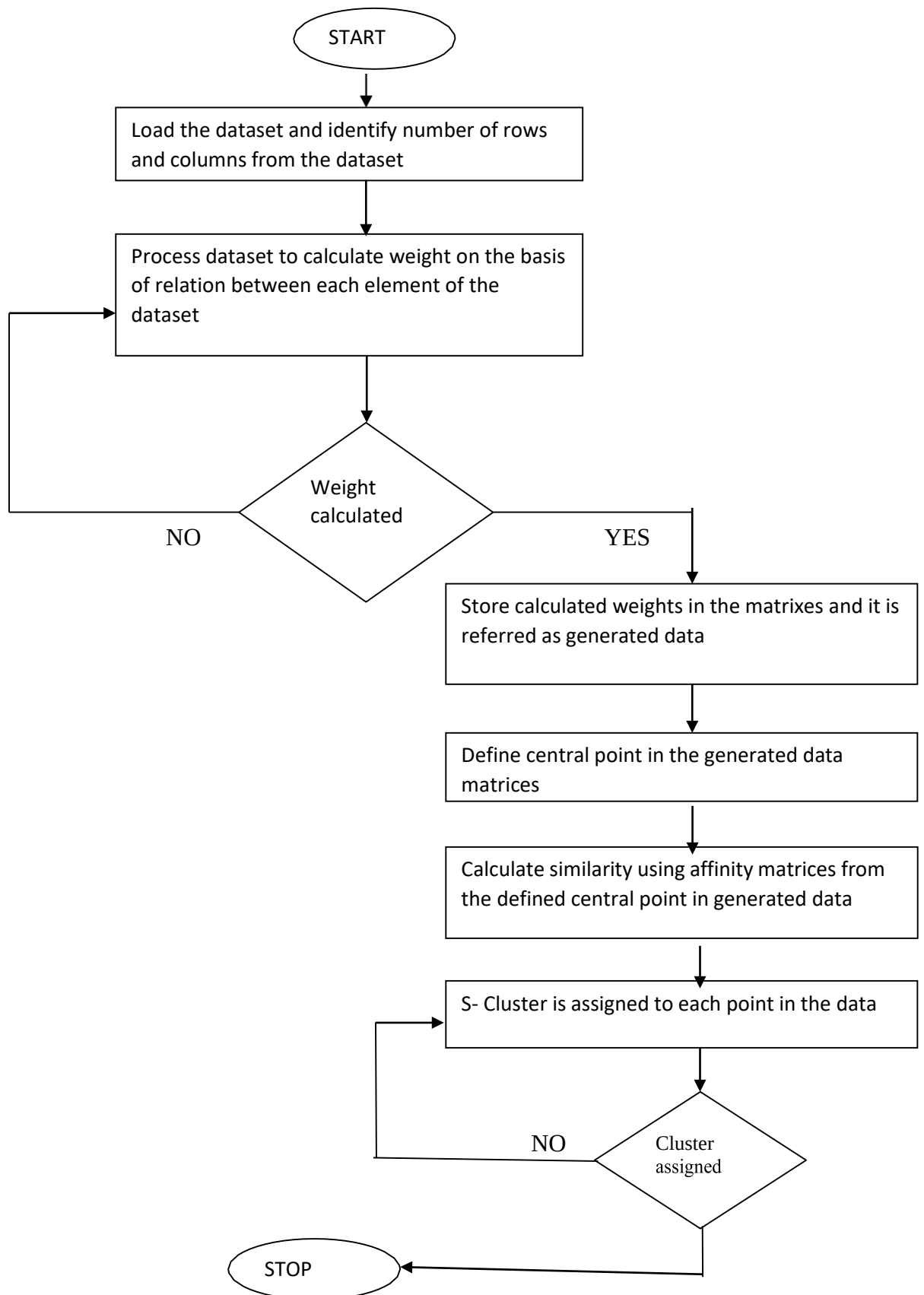


Figure3.1: Flow chart of proposed algorithm

3.5 Description of Existing Algorithm

In the existing algorithm technique of affinity matrices is applied calculate similarity between the elements. In the affinity matrices the relationship between the elements is calculated which are adjacent to each other. The elements which have higher relationship have higher weights. The elements are arranged in the descending order of similarity. But the result was not effective as much because its noise is high along its accuracy percentage is low.

3.6 Description of Proposed Algorithm

In this Proposed algorithm technique of mean shift is applied calculate the similarity between the elements which is present inside matrix. In this algorithm relationship between the elements is calculated which are one end to another end, so that it better way to find the relationship among all the elements present in the matrix. The elements are arranged according to its similarity in descending order. So that result was enhanced and its overall performance is increased. So the result was effective and its noise is low along its accuracy percentage is gets increased.

CHAPTER 4

RESULTS AND DISCUSSION

The algorithm is implemented using MATLAB 7.13 programming language. This algorithm is tested on the Datasets. In this experiment, we have tested the algorithm on three Datasets. We compared our results of Existing algorithm with proposed algorithm. This comparison is made using Time, Noise, and Accuracy Percentage. The values for all of these criteria are calculated and then the final results are shown in a table and also graphs are plotted to show the comparison between both of the approaches of Clusters data. Lastly, the final cluster is assigned to each element of the data by using of proposed algorithm.

Euclidean Distance:-

$$\sqrt{(x_1 - x)^2 + (y_1 - y)^2}$$

Accuracy Percentage:-

$$\frac{\text{No. of Point clustered}}{\text{Total No. of Points}} \times 100$$

Noise:-

$$(\text{Total No. of Points}) - (\text{No. of Point Clustered})$$

4.1 EXPERIMENTAL RESULTS

The Optimized Data cluster based on Asymmetric algorithm and K-means are implemented on three data sets Iris, Breast Cancer, Bank. It is finding similarity between all the elements of data using affinity metrics. The results are stored in descending order so that it will produce better result for assigning cluster, but the result was not effective as much expected once and its noise is also high and its accuracy percentage is very low.

Results of Existing algorithm are:-

Clustered data using K-means clustering for Iris data set shown in figure (4.1), for Breast cancer data set shown in figure (4.3), for Bank data set shown in figure (4.5). Clustered data using Asymmetric clustering for Iris data set shown in figure (4.2), for Breast cancer data set shown in figure (4.4), for Bank data set shown in figure (4.6).

Results of Proposed algorithm are:-

Clustered data using K-means clustering and Normalization form for Iris data set shown in figure (4.8), for Breast cancer data set shown in figure (4.11), for Bank data set shown in figure (4.14), Point set of Iris data in figure (4.7), Point set of Breast cancer data in figure (4.10), Point set of Bank data in figure (4.13) Clustered data using Asymmetric clustering form for Iris data set shown in figure (4.9), for Breast cancer data set shown in figure (4.12), for Bank data set shown in figure (4.15).



Figure 4.1: K-Means clustering of Iris data

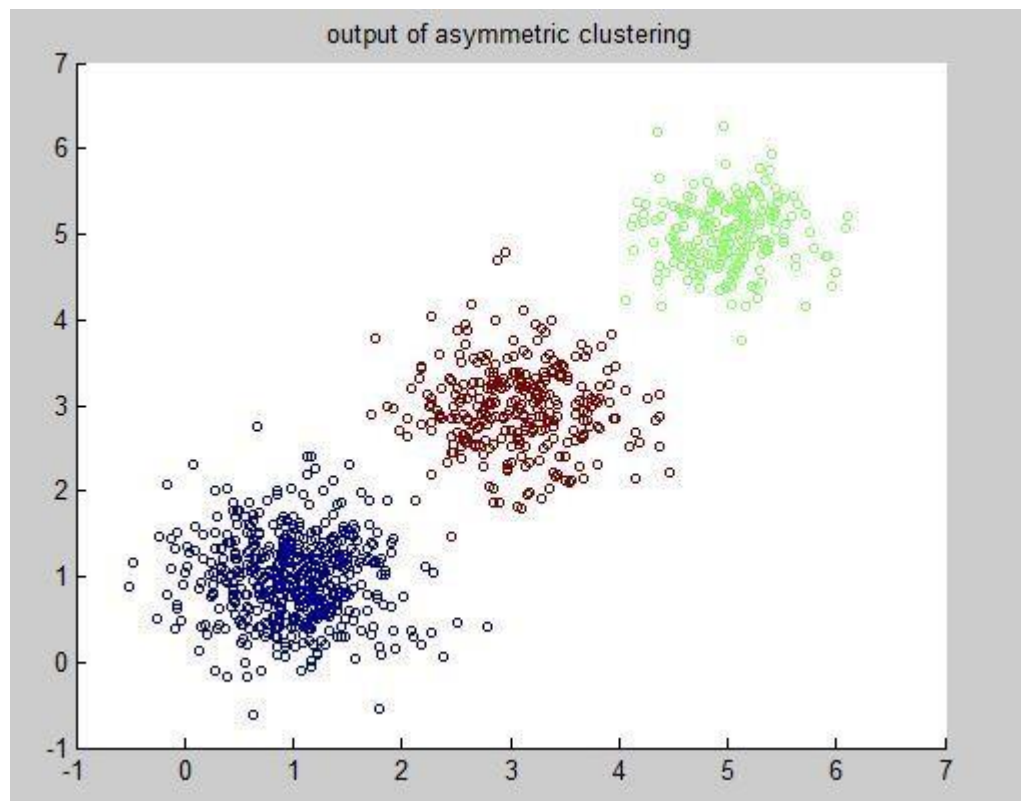


Figure 4.2: Asymmetric clustering of Iris data



Figure 4.3: K-means clustering of Brest cancer data

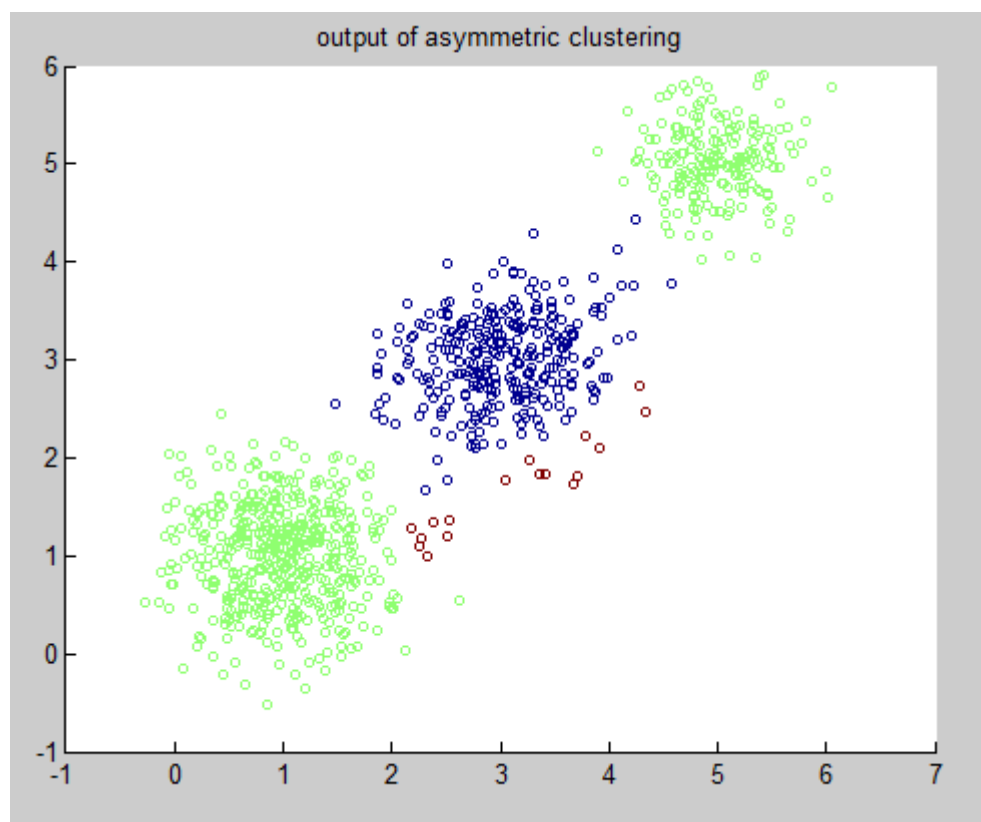


Figure 4.4: Asymmetric clustering of Breast cancer data

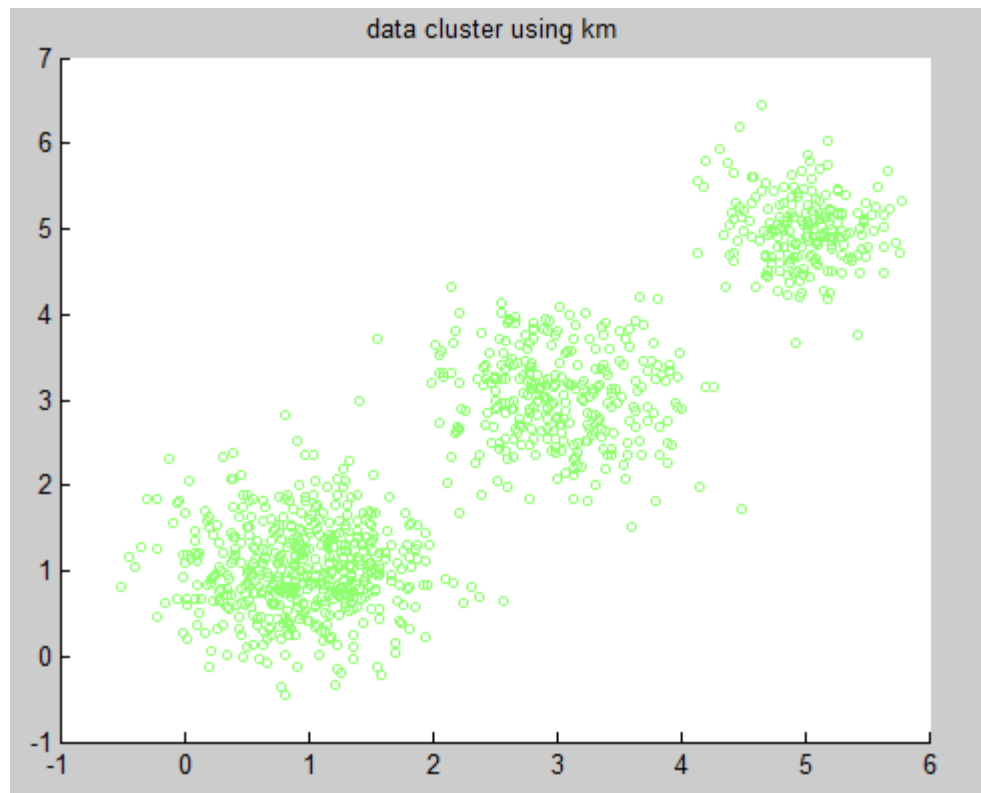


Figure 4.5: K-means clustering of Bank data

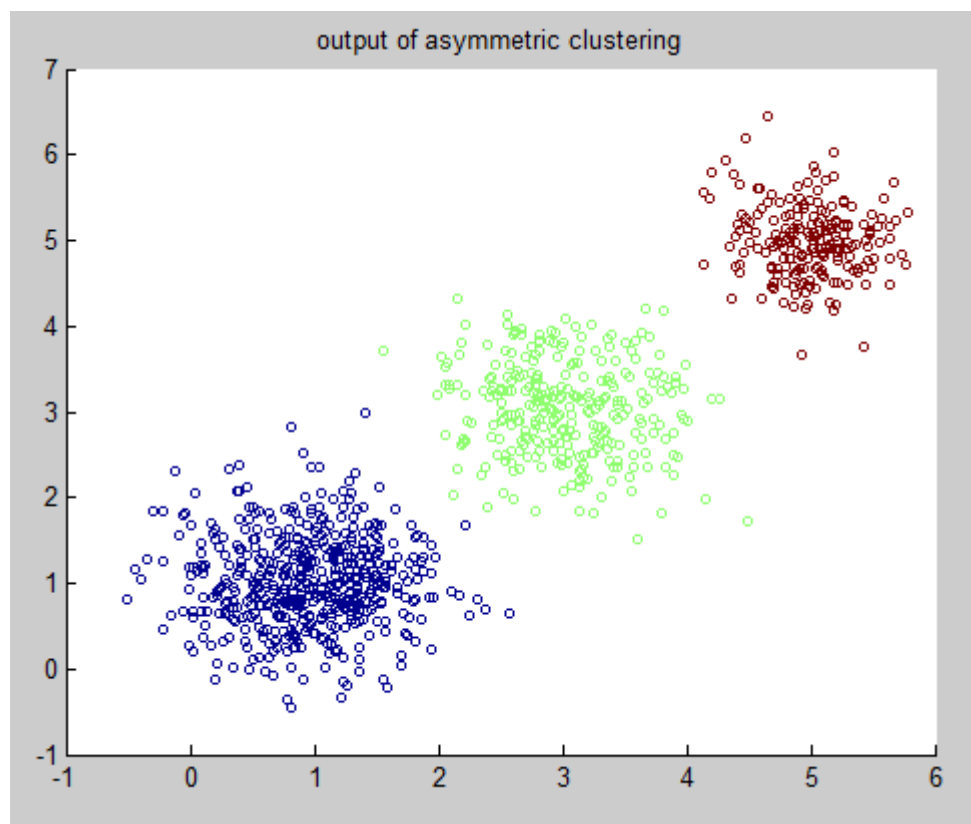


Figure 4.6: Asymmetric clustering of Bank data

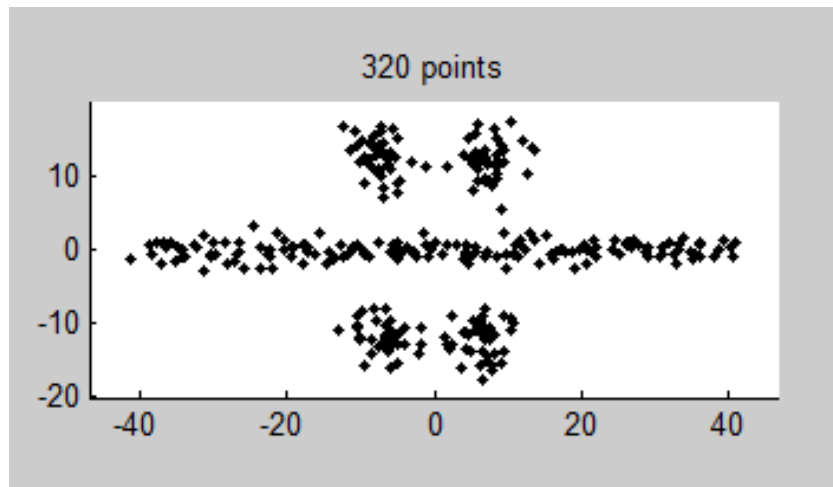


Figure 4.7: Point sets of Iris data

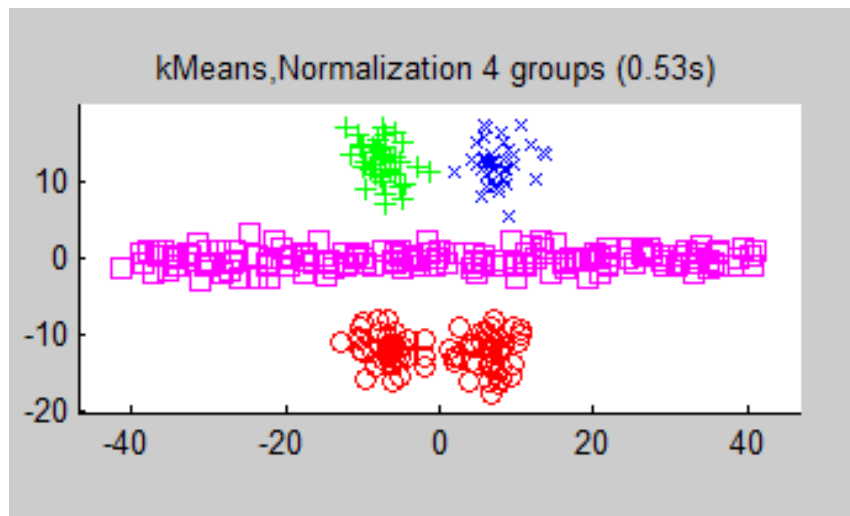


Figure 4.8: k-means normalized group of Iris data

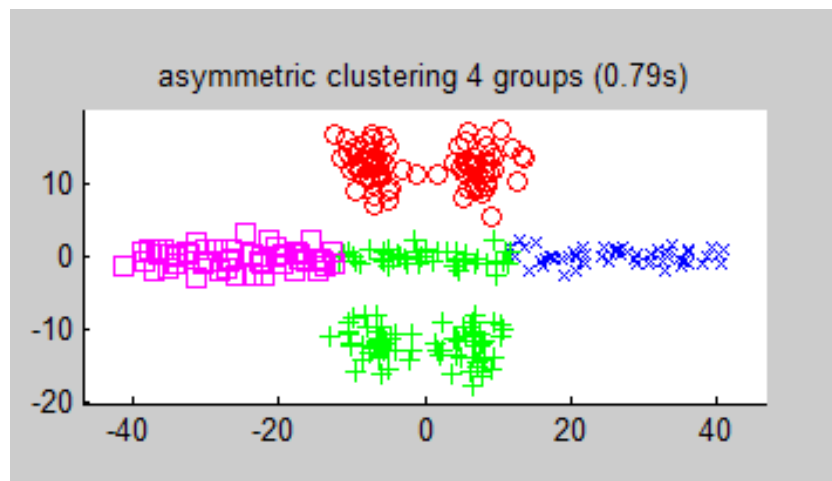


Figure 4.9: Asymmetric clustering of Iris data

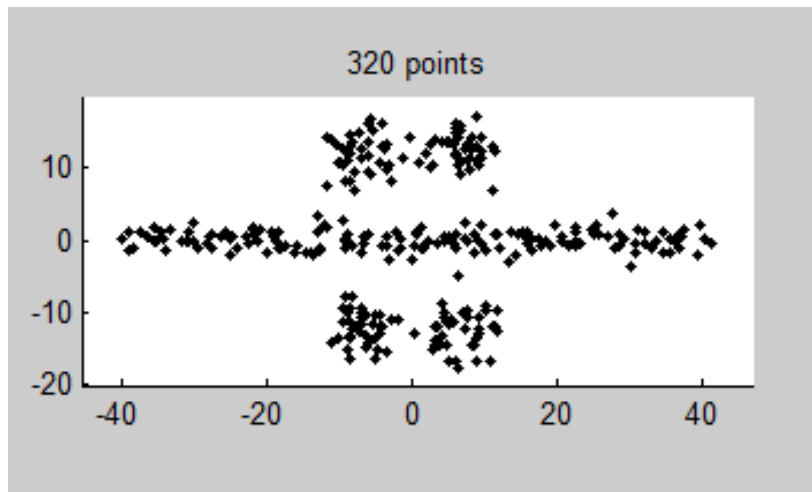


Figure 4.10: Point set of Brest Cancer data

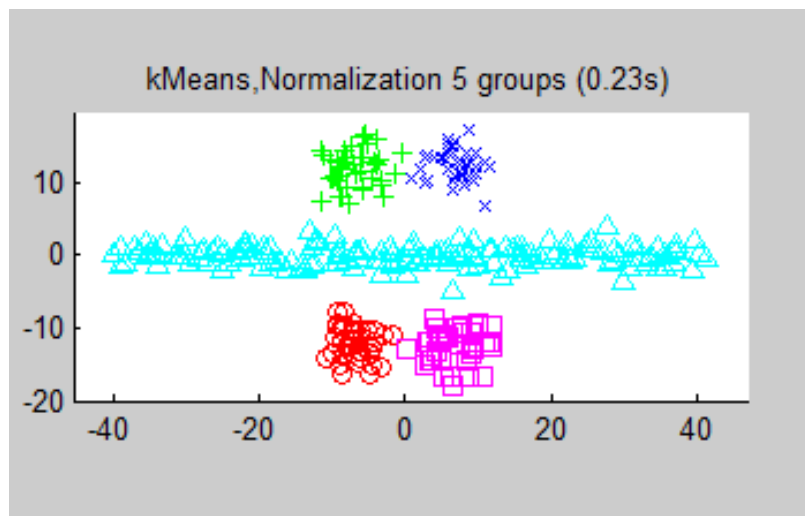


Figure 4.11: k-means normalized group of Breast cancer data

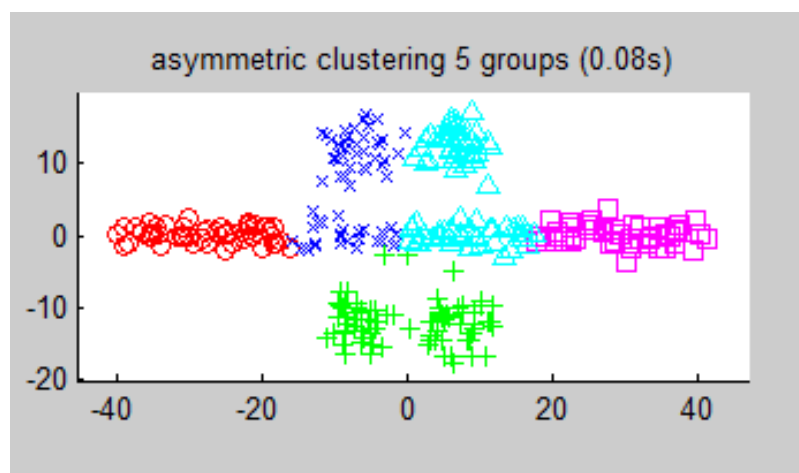


Figure 4.12: Asymmetric clustering of Breast cancer data

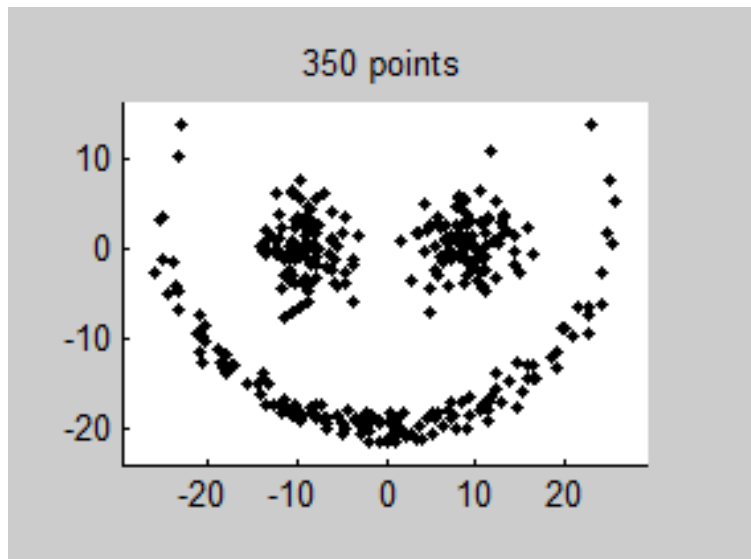


Figure 4.13: Point set of Bank data

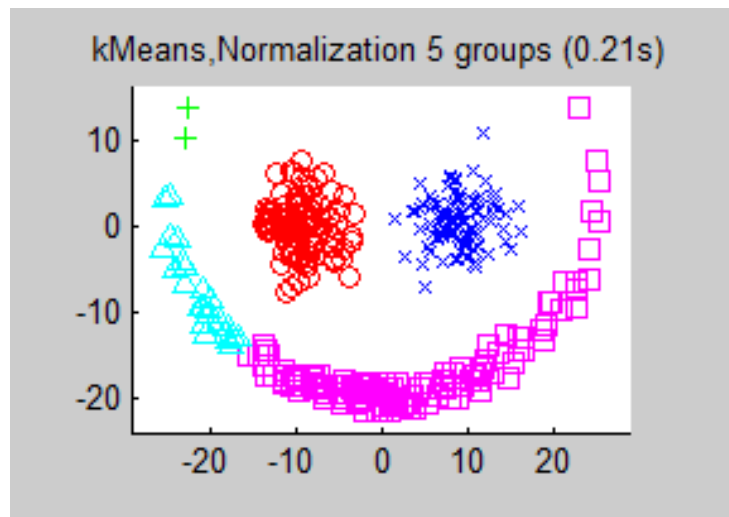


Figure 4.14: k-means normalized group of Bank data

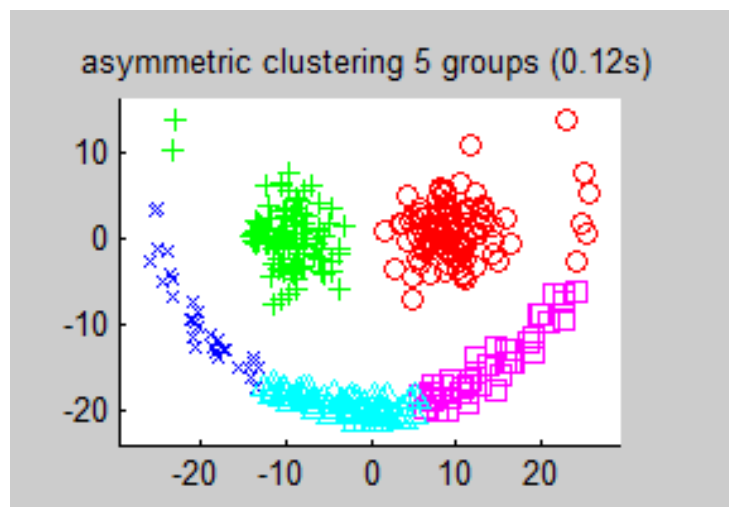


Figure 4.15: Asymmetric clustering of Bank data

4.2 COMPARISION WITH EXISTING TECHNIQUE

The comparison between the existing and new technique is shown in the form of table and graphs. Firstly the table is shown and then the three graphs are shown. To represent tables and graphs are plotted below in this section.

Results of Comparison between Existing Algorithm and Proposed Algorithm are shown in table (4.1).

Graphical representation of Comparison between Existing Algorithm and Proposed Algorithm are shown in figure.

For Iris data figure used (4.16), for Breast cancer data figure used (4.17), for Bank data figure used (4.18)

Table 4.1: The result of the experiment

Problem	Algorithm	Time	Noise	Accuracy Percentage
Iris	EXISTING	0.99771	0.998111	57.14%
	PROPOSED	1.00172	0	92.86%
Breast Cancer	EXISTING	1.31546	0.994359	64.29%
	PROPOSED	1.19809	0	92.86%
Bank	EXISTING	1.40702	1.00365	71.43%
	PROPOSED	1.10306	0.015625	93.33%

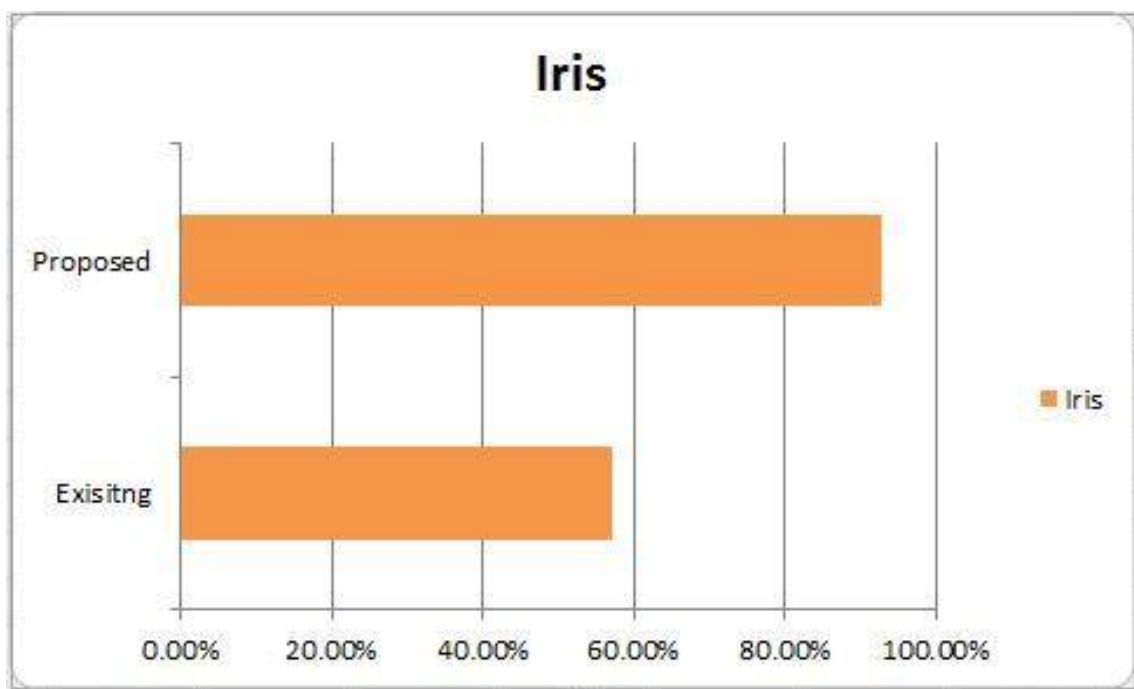


Figure 4.16: Iris data

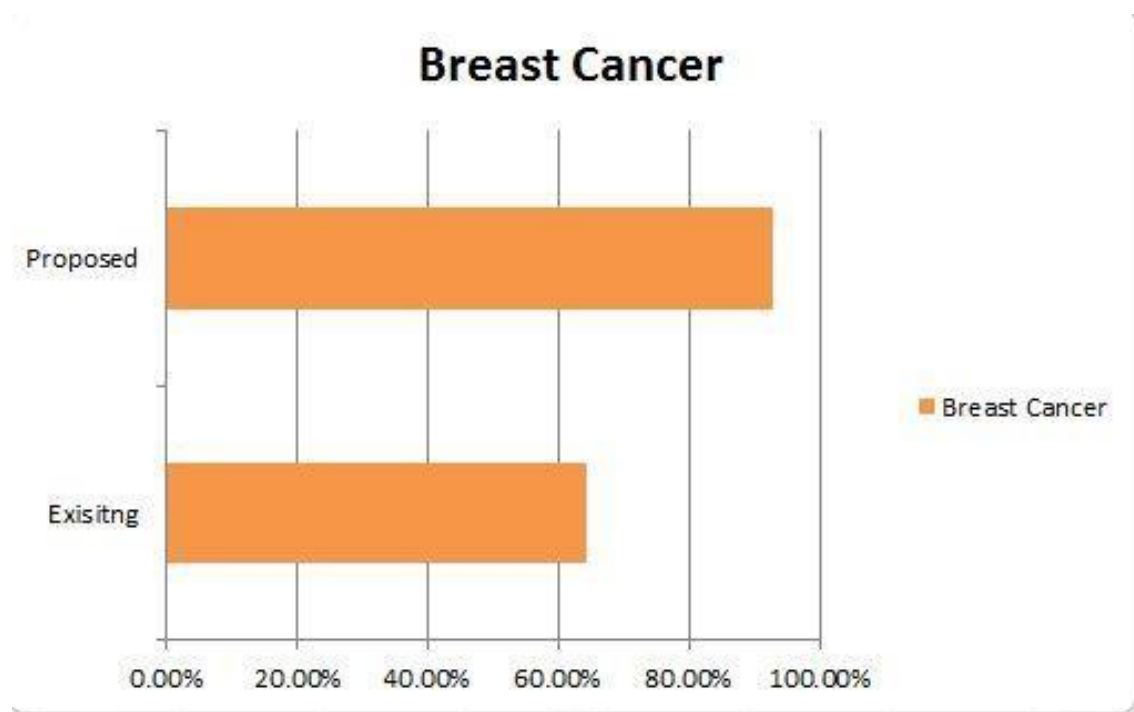


Figure 4.17: Breast cancer data

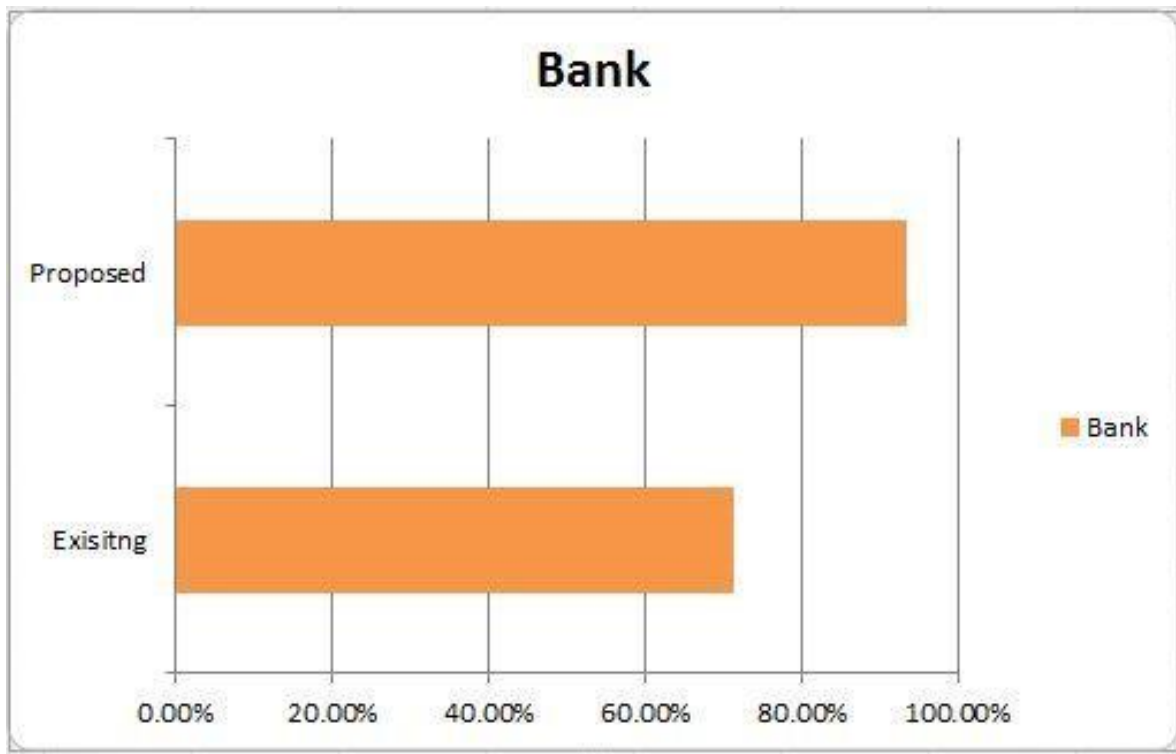


Figure 4.18: Bank data

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

The whole study and all the sections of this thesis report can be concluded as follows. In first section, all the concepts that are used in this work are explained briefly. It includes data clustering, the clustering techniques, k-means clustering, asymmetric clustering and s-clustering. It includes clustered data, the optimization techniques, affinity metrics algorithm and mean-shift algorithm. In second section, the literature review of all the papers studied is discussed. Further in next section, the present work, the methodology, the objectives of the study and the flow chart is explained. Lastly, the experimental results of the study are analyzed and discussed. It also includes the comparison between already existing and new approach.

The whole study and thesis work is concluded in the following section. There are still chances of improvement in the existing work and this is discussed in the future work section.

5.1 CONCLUSION

Data Clustering is the most important and crucial process for identifying the data. It has several applications and among them most important is Clustered of Data. In case of huge amount of data set, clustering is most important. So, we require a technique that can be used to attain data should be in arranged in well-mannered cluster. There are many techniques that can be used for this purpose. Clustering method for data clustering gives the promising results. Hence, I have used clustering techniques for clustering the data.

In my work, I have surveyed PSO algorithm and different variations of this algorithm are analyzed. The variations of PSO contribute that it finds similarity between its nearby neighbors only. So we apply affinity metrics for solve this because it finds the similarity between adjacent elements of matrix, but the result was not effective as much as expected. At last finally apply mean shift algorithm for overcoming from

this problem. It provides better techniques from all old one and it improves the results also. The results of clustered data is in descending order and increase the accuracy percentage and also reduce the noise.

5.2 FUTURE SCOPE

There is scope of betterment of this algorithm. Firstly, the computational time of this method is not as much reduces from the older once. So, we can try to reduce this computational time in future work. Secondly, in this method we have used mean shift for optimization purpose. There are many other techniques too like Genetic algorithm search can be also used instead of mean shift Algorithm and the results of both techniques can be compared and the one that produces better results can be introduced in the method.

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APPENDIX

A

Association

Asymmetric Clustering Algorithm

Ant Colony optimization

Affinity Metric

B

Big Data

C

Classification

D

Data Clustering

Data Mining

Density-based Clustering Algorithm

E

Estimation

F

Fuzzy C-Means Algorithm

G

Grid-based Clustering Algorithm

Genetic Algorithm

H

Hierarchical-based Clustering

I

Image Segmentation

K

K-Means Clustering Algorithm

M

Mean Shift Algorithm

Markov Clustering algorithm

P

Partitioning Algorithm

Particle Swarm Optimization

S

Spectral Clustering